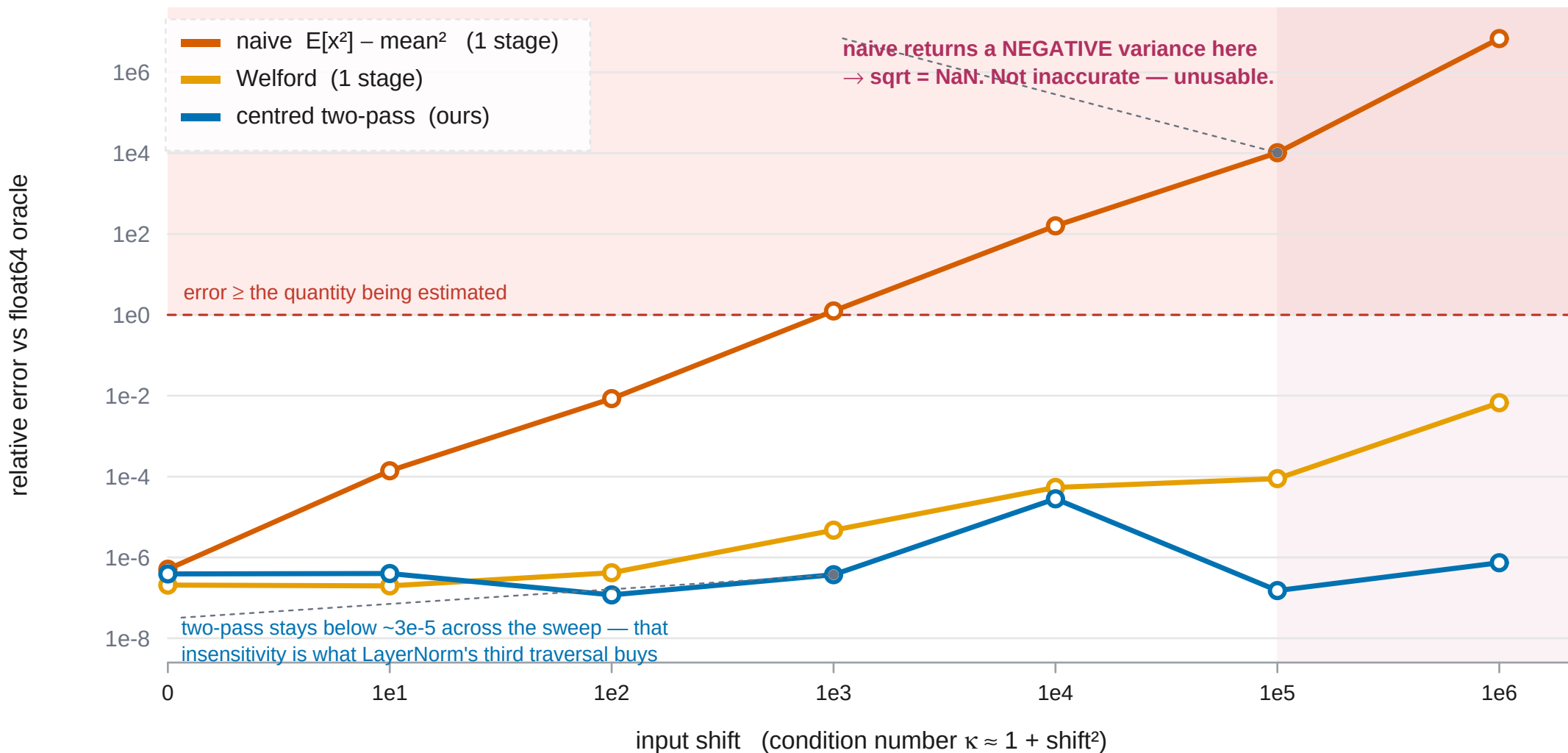


Where each FP32 variance formulation breaks

N = 512. Shifting the data leaves the variance unchanged but scales the conditioning by $\sim \text{shift}^2$.



RMSNorm never forms a mean, so it never performs this subtraction: its error stays below $\sim 1.8e-5$ across the sweep.